2021/ICT/51

IT1223 (P) - Database Management Systems (P)
Department of Physical Science
University of Vavuniya
Revision Work Sheet - 01

1. Create database **'revision_worksheet'**.

   CODING:

   CREATE databases **revision_worksheet;**

   USE **revision_worksheet;**

2. Create tables students, courses and enrollment with suitable datatypes.

   CODING:

```
CREATE TABLE students (
    student_id INT PRIMARY KEY,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    age INT,
    gender VARCHAR(10),
    city VARCHAR(50)
);
```

   OUTPUT:

```
MariaDB [(none)]> use revision_worksheet;
Database changed
MariaDB [revision_worksheet]> CREATE TABLE students (
    ->     student_id INT PRIMARY KEY,
    ->     first_name VARCHAR(50),
    ->     last_name VARCHAR(50),
    ->     age INT,
    ->     gender VARCHAR(10),
    ->     city VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.011 sec)
```

```
CREATE TABLE courses (
    course_id INT PRIMARY KEY,
    course_name VARCHAR(100),
    instructor VARCHAR(50),
```

```
    credits INT,
    department VARCHAR(50)
);
```

OUTPUT:

```
MariaDB [revision_worksheet]> CREATE TABLE courses (
    ->     course_id INT PRIMARY KEY,
    ->     course_name VARCHAR(100),
    ->     instructor VARCHAR(50),
    ->     credits INT,
    ->     department VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.016 sec)
```

```
CREATE TABLE enrollment (
    enrollment_id INT PRIMARY KEY,
    student_id INT,
    course_id INT,
    enrollment_date DATE,
    FOREIGN KEY (student_id) REFERENCES students(student_id),
    FOREIGN KEY (course_id) REFERENCES courses(course_id)
);
```

OUTPUT:

```
MariaDB [revision_worksheet]> CREATE TABLE enrollment (
    ->     enrollment_id INT PRIMARY KEY,
    ->     student_id INT,
    ->     course_id INT,
    ->     enrollment_date DATE,
    ->     FOREIGN KEY (student_id) REFERENCES students(student_id),
    ->     FOREIGN KEY (course_id) REFERENCES courses(course_id)
    -> );
Query OK, 0 rows affected (0.019 sec)
```

3. Write a SQL query to view the structure of the **students**, **courses** and **enrollment** tables.

CODING:

```
DESCRIBE students;
```

OUTPUT:

```
MariaDB [revision_worksheet]> DESCRIBE students;
+------------+------------+------+-----+---------+-------+
| Field      | Type       | Null | Key | Default | Extra |
+------------+------------+------+-----+---------+-------+
| student_id | int(11)    | NO   | PRI | NULL    |       |
| first_name | varchar(50)| YES  |     | NULL    |       |
| last_name  | varchar(50)| YES  |     | NULL    |       |
| age        | int(11)    | YES  |     | NULL    |       |
| gender     | varchar(10)| YES  |     | NULL    |       |
| city       | varchar(50)| YES  |     | NULL    |       |
+------------+------------+------+-----+---------+-------+
6 rows in set (0.008 sec)
```

CODING:

DESCRIBE courses;

OUTPUT:

```
MariaDB [revision_worksheet]> DESCRIBE courses;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| course_id   | int(11)     | NO   | PRI | NULL    |       |
| course_name | varchar(100)| YES  |     | NULL    |       |
| instructor  | varchar(50) | YES  |     | NULL    |       |
| credits     | int(11)     | YES  |     | NULL    |       |
| department  | varchar(50) | YES  |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
5 rows in set (0.008 sec)
```

CODING:

DESCRIBE enrollment;

OUTPUT:

```
MariaDB [revision_worksheet]> DESCRIBE enrollment;
+-----------------+---------+------+-----+---------+-------+
| Field           | Type    | Null | Key | Default | Extra |
+-----------------+---------+------+-----+---------+-------+
| enrollment_id   | int(11) | NO   | PRI | NULL    |       |
| student_id      | int(11) | YES  | MUL | NULL    |       |
| course_id       | int(11) | YES  | MUL | NULL    |       |
| enrollment_date | date    | YES  |     | NULL    |       |
+-----------------+---------+------+-----+---------+-------+
4 rows in set (0.006 sec)
```

4. How to insert the values into table **students**, **courses** and **enrollment** tables.

CODING:

*** Inserting into students table**

INSERT INTO students (student_id, first_name, last_name, age, gender, city) VALUES

(1, 'John', 'Doe', 20, 'Male', 'New York'),

(2, 'Jane', 'Smith', 22, 'Female', 'Los Angeles'),

(3, 'Michael', 'Johnson', 21, 'Male', 'Chicago'),

(4, 'Emily', 'Brown', 19, 'Female', 'Houston'),

(5, 'David', 'Martinez', 23, 'Male', 'Miami'),

(6, 'Sarah', 'Jones', 20, 'Female', 'Boston'),

(7, 'Daniel', 'Garcia', 21, 'Male', 'Seattle'),

(8, 'Jennifer', 'Wilson', 22, 'Female', 'San Francisco'),

(9, 'Christopher', 'Anderson', 24, 'Male', 'Dallas'),

(10, 'Amanda', 'Thomas', 19, 'Female', 'Phoenix');

OUTPUT:

```
MariaDB [revision_worksheet]> INSERT INTO students (student_id, first_name
last_name, age, gender, city) VALUES
    -> (1, 'John', 'Doe', 20, 'Male', 'New York'),
    -> (2, 'Jane', 'Smith', 22, 'Female', 'Los Angeles'),
    -> (3, 'Michael', 'Johnson', 21, 'Male', 'Chicago'),
    -> (4, 'Emily', 'Brown', 19, 'Female', 'Houston'),
    -> (5, 'David', 'Martinez', 23, 'Male', 'Miami'),
    -> (6, 'Sarah', 'Jones', 20, 'Female', 'Boston'),
    -> (7, 'Daniel', 'Garcia', 21, 'Male', 'Seattle'),
    -> (8, 'Jennifer', 'Wilson', 22, 'Female', 'San Francisco'),
    -> (9, 'Christopher', 'Anderson', 24, 'Male', 'Dallas'),
    -> (10, 'Amanda', 'Thomas', 19, 'Female', 'Phoenix');
Query OK, 10 rows affected (0.011 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

CODING:

***Inserting into courses table**

INSERT INTO courses (course_id, course_name, instructor, credits, department) VALUES

(1, 'Introduction to Programming', 'Prof. Smith', 3, 'Computer Science'),

(2, 'Database Management', 'Prof. Johnson', 4, 'Computer Science'),

(3, 'Statistics', 'Prof. Brown', 3, 'Mathematics'),

(4, 'English Composition', 'Prof. Martinez', 3, 'English'),

(5, 'History of Art', 'Prof. Wilson', 3, 'Art'),

(6, 'Chemistry', 'Prof. White', 4, 'Science'),

(7, 'Economics', 'Prof. Garcia', 3, 'Social Science'),

(8, 'Algebra', 'Prof. Thompson', 3, 'Mathematics'),

(9, 'Psychology', 'Prof. Davis', 3, 'Social Science'),

(10, 'Physical Education', 'Prof. Lee', 2, 'Physical Education');

OUTPUT:

```
MariaDB [revision_worksheet]> INSERT INTO courses (course_id, course_name,
nstructor, credits, department) VALUES
    -> (1, 'Introduction to Programming', 'Prof. Smith', 3, 'Computer Scie
e'),
    -> (2, 'Database Management', 'Prof. Johnson', 4, 'Computer Science'),
    -> (3, 'Statistics', 'Prof. Brown', 3, 'Mathematics'),
    -> (4, 'English Composition', 'Prof. Martinez', 3, 'English'),
    -> (5, 'History of Art', 'Prof. Wilson', 3, 'Art'),
    -> (6, 'Chemistry', 'Prof. White', 4, 'Science'),
    -> (7, 'Economics', 'Prof. Garcia', 3, 'Social Science'),
    -> (8, 'Algebra', 'Prof. Thompson', 3, 'Mathematics'),
    -> (9, 'Psychology', 'Prof. Davis', 3, 'Social Science'),
    -> (10, 'Physical Education', 'Prof. Lee', 2, 'Physical Education');
Query OK, 10 rows affected (0.002 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

CODING:

*** Inserting into enrollment table**

```
INSERT INTO enrollment (enrollment_id, student_id, course_id, enrollment_date) VALUES
(1, 1, 1, '2024-01-15'),
(2, 1, 3, '2024-01-20'),
(3, 2, 2, '2024-02-01'),
(4, 3, 4, '2024-02-10'),
(5, 4, 5, '2024-03-05'),
(6, 5, 6, '2024-03-10'),
(7, 6, 7, '2024-03-15'),
(8, 7, 8, '2024-04-01'),
(9, 8, 9, '2024-04-05'),
(10, 9, 10, '2024-04-10');
```

OUTPUT:

```
MariaDB [revision_worksheet]> INSERT INTO enrollment (enrollment_id, studen
_id, course_id, enrollment_date) VALUES
    -> (1, 1, 1, '2024-01-15'),
    -> (2, 1, 3, '2024-01-20'),
    -> (3, 2, 2, '2024-02-01'),
    -> (4, 3, 4, '2024-02-10'),
    -> (5, 4, 5, '2024-03-05'),
    -> (6, 5, 6, '2024-03-10'),
    -> (7, 6, 7, '2024-03-15'),
    -> (8, 7, 8, '2024-04-01'),
    -> (9, 8, 9, '2024-04-05'),
    -> (10, 9, 10, '2024-04-10');
Query OK, 10 rows affected (0.003 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

5. Retrieve the students details who are older than 20.

   CODING:

   SELECT * FROM students WHERE age > 20;

   OUTPUT:

```
MariaDB [revision_worksheet]> SELECT * FROM students WHERE age > 20;
+------------+-------------+-----------+------+--------+---------------
| student_id | first_name  | last_name | age  | gender | city
+------------+-------------+-----------+------+--------+---------------
|          2 | Jane        | Smith     |   22 | Female | Los Angeles
|          3 | Michael     | Johnson   |   21 | Male   | Chicago
|          5 | David       | Martinez  |   23 | Male   | Miami
|          7 | Daniel      | Garcia    |   21 | Male   | Seattle
|          8 | Jennifer    | Wilson    |   22 | Female | San Francisco
|          9 | Christopher | Anderson  |   24 | Male   | Dallas
+------------+-------------+-----------+------+--------+---------------
6 rows in set (0.001 sec)
```

6. Retrieve students who are older than 20 and from New York, or younger than 20 and from Los Angeles.

   CODING:

   SELECT * FROM students WHERE (age > 20 AND city = 'New York') OR (age < 20 AND city = 'Los Angeles');

   OUTPUT:

```
MariaDB [revision_worksheet]> SELECT * FROM students WHERE (age > 20 AND c
y = 'New York') OR (age < 20 AND city = 'Los Angeles');
Empty set (0.002 sec)
```

7. Retrieve students who are not from New York.

   CODING:

   SELECT * FROM students WHERE city <> 'New York';

   OUTPUT:

```
MariaDB [revision_worksheet]> SELECT * FROM students WHERE city <> 'New Yor
';
+------------+-------------+-----------+------+--------+---------------+
| student_id | first_name  | last_name | age  | gender | city          |
+------------+-------------+-----------+------+--------+---------------+
|          2 | Jane        | Smith     |   22 | Female | Los Angeles   |
|          3 | Michael     | Johnson   |   21 | Male   | Chicago       |
|          4 | Emily       | Brown     |   19 | Female | Houston       |
|          5 | David       | Martinez  |   23 | Male   | Miami         |
|          6 | Sarah       | Jones     |   20 | Female | Boston        |
|          7 | Daniel      | Garcia    |   21 | Male   | Seattle       |
|          8 | Jennifer    | Wilson    |   22 | Female | San Francisco |
|          9 | Christopher | Anderson  |   24 | Male   | Dallas        |
|         10 | Amanda      | Thomas    |   19 | Female | Phoenix       |
+------------+-------------+-----------+------+--------+---------------+
9 rows in set (0.001 sec)
```

8. Write a SQL query to find the youngest student’s age and oldest student's age.

   CODING:

   SELECT MIN(age) AS youngest_age, MAX(age) AS oldest_age FROM students;

   OUTPUT:

```
MariaDB [revision_worksheet]> SELECT MIN(age) AS youngest_age, MAX(age) AS o
ldest_age FROM students;
+--------------+------------+
| youngest_age | oldest_age |
+--------------+------------+
|           19 |         24 |
+--------------+------------+
1 row in set (0.001 sec)
```

9. Find the both the sum and the average of credits in the courses table using aggregate functions.

   CODING:

   SELECT SUM(credits) AS total_credits, AVG(credits) AS average_credits FROM courses;

   OUTPUT:

```
MariaDB [revision_worksheet]> SELECT SUM(credits) AS total_credits, AVG(cred
its) AS average_credits FROM courses;
+---------------+-----------------+
| total_credits | average_credits |
+---------------+-----------------+
|            31 |          3.1000 |
+---------------+-----------------+
1 row in set (0.001 sec)
```

10. Count the number of female students in the students table.

    CODING:

    SELECT COUNT(*) AS female_students_count FROM students WHERE gender = 'Female';

OUTPUT:

```
MariaDB [revision_worksheet]> SELECT COUNT(*) AS female_students_count FRO
students WHERE gender = 'Female';
+-----------------------+
| female_students_count |
+-----------------------+
|                     5 |
+-----------------------+
1 row in set (0.000 sec)
```

11. Retrieve the first 3 students from the students table.

    CODING:

    SELECT * FROM students LIMIT 3;

    OUTPUT:

```
MariaDB [revision_worksheet]> SELECT * FROM students LIMIT 3;
+------------+------------+-----------+------+--------+-------------+
| student_id | first_name | last_name | age  | gender | city        |
+------------+------------+-----------+------+--------+-------------+
|          1 | John       | Doe       |   20 | Male   | New York    |
|          2 | Jane       | Smith     |   22 | Female | Los Angeles |
|          3 | Michael    | Johnson   |   21 | Male   | Chicago     |
+------------+------------+-----------+------+--------+-------------+
3 rows in set (0.001 sec)
```

12. Write a SQL query to find the students who are from New York and Los Angeles. (Hint: Use IN operator)

    CODING:

    SELECT * FROM students WHERE city IN ('New York', 'Los Angeles');

    OUTPUT:

```
MariaDB [revision_worksheet]> SELECT * FROM students WHERE city IN ('New Yo
k', 'Los Angeles');
+------------+------------+-----------+------+--------+-------------+
| student_id | first_name | last_name | age  | gender | city        |
+------------+------------+-----------+------+--------+-------------+
|          1 | John       | Doe       |   20 | Male   | New York    |
|          2 | Jane       | Smith     |   22 | Female | Los Angeles |
+------------+------------+-----------+------+--------+-------------+
2 rows in set (0.001 sec)
```

13. Retrieve all courses with a credit count between 3 and 4.

    CODING:

    SELECT * FROM courses WHERE credits BETWEEN 3 AND 4;

OUTPUT:

```
MariaDB [revision_worksheet]> SELECT * FROM courses WHERE credits BETWEEN 3 AND 4;
+-----------+---------------------------+---------------+---------+-----------------
| course_id | course_name               | instructor    | credits | department
+-----------+---------------------------+---------------+---------+-----------------
|         1 | Introduction to Programming | Prof. Smith   |       3 | Computer Science
|         2 | Database Management       | Prof. Johnson |       4 | Computer Science
|         3 | Statistics                | Prof. Brown   |       3 | Mathematics
|         4 | English Composition       | Prof. Martinez |      3 | English
|         5 | History of Art            | Prof. Wilson  |       3 | Art
|         6 | Chemistry                 | Prof. White   |       4 | Science
|         7 | Economics                 | Prof. Garcia  |       3 | Social Science
|         8 | Algebra                   | Prof. Thompson |      3 | Mathematics
|         9 | Psychology                | Prof. Davis   |       3 | Social Science
+-----------+---------------------------+---------------+---------+-----------------
9 rows in set (0.001 sec)
```

14. Find the course details with more than 3 credits.

CODING:

SELECT * FROM courses WHERE credits > 3;

OUTPUT:

```
MariaDB [revision_worksheet]> SELECT * FROM courses WHERE credits > 3;
+-----------+---------------------+---------------+---------+-----------------
| course_id | course_name         | instructor    | credits | department
+-----------+---------------------+---------------+---------+-----------------
|         2 | Database Management | Prof. Johnson |       4 | Computer Science
|         6 | Chemistry           | Prof. White   |       4 | Science
+-----------+---------------------+---------------+---------+-----------------
2 rows in set (0.001 sec)
```

15. Write a SQL query to count the number of students in each city.

CODING:

SELECT city, COUNT(*) AS number_of_students FROM students GROUP BY city;

OUTPUT:

```
MariaDB [revision_worksheet]> SELECT city, COUNT(*) AS number_of_students FROM students GROUP BY city
+---------------+--------------------+
| city          | number_of_students |
+---------------+--------------------+
| Boston        |                  1 |
| Chicago       |                  1 |
| Dallas        |                  1 |
| Houston       |                  1 |
| Los Angeles   |                  1 |
| Miami         |                  1 |
| New York      |                  1 |
| Phoenix       |                  1 |
| San Francisco |                  1 |
| Seattle       |                  1 |
+---------------+--------------------+
10 rows in set (0.001 sec)
```

16. Find the cities that start with ‘S’.

    CODING:

    SELECT DISTINCT city FROM students WHERE city LIKE 'S%';

    OUTPUT:

```
MariaDB [revision_worksheet]> SELECT DISTINCT city FROM students WHERE city
LIKE 'S%';
+---------------+
| city          |
+---------------+
| Seattle       |
| San Francisco |
+---------------+
2 rows in set (0.001 sec)
```

17. Retrieve courses whose name contains 'Programming' or 'Management'.

    CODING:

    SELECT * FROM courses WHERE course_name LIKE '%Programming%' OR course_name LIKE '%Management%';

    OUTPUT:

```
MariaDB [revision_worksheet]> SELECT * FROM courses WHERE course_name LIKE '%Programming%' OR course_name LIKE '%Management%';
+-----------+----------------------------+---------------+---------+------------------+
| course_id | course_name                | instructor    | credits | department       |
+-----------+----------------------------+---------------+---------+------------------+
|         1 | Introduction to Programming | Prof. Smith   |       3 | Computer Science |
|         2 | Database Management        | Prof. Johnson |       4 | Computer Science |
+-----------+----------------------------+---------------+---------+------------------+
2 rows in set (0.001 sec)
```

18. How to retrieve students enrolled in courses using MYSQL.

    CODING:

    SELECT students.student_id, students.first_name, students.last_name, courses.course_name

    FROM enrollment

    JOIN students ON enrollment.student_id = students.student_id

    JOIN courses ON enrollment.course_id = courses.course_id;

    OUTPUT:

```
MariaDB [revision_worksheet]> SELECT students.student_id, students.first_name, students.last_name, courses.course_n
    -> FROM enrollment
    -> JOIN students ON enrollment.student_id = students.student_id
    -> JOIN courses ON enrollment.course_id = courses.course_id;
+------------+-------------+-----------+-----------------------------+
| student_id | first_name  | last_name | course_name                 |
+------------+-------------+-----------+-----------------------------+
|          1 | John        | Doe       | Introduction to Programming |
|          1 | John        | Doe       | Statistics                  |
|          2 | Jane        | Smith     | Database Management         |
|          3 | Michael     | Johnson   | English Composition         |
|          4 | Emily       | Brown     | History of Art              |
|          5 | David       | Martinez  | Chemistry                   |
|          6 | Sarah       | Jones     | Economics                   |
|          7 | Daniel      | Garcia    | Algebra                     |
|          8 | Jennifer    | Wilson    | Psychology                  |
|          9 | Christopher | Anderson  | Physical Education          |
+------------+-------------+-----------+-----------------------------+
10 rows in set (0.001 sec)
```

19. Retrieve courses along with students enrolled in them.

    CODING:

    SELECT courses.course_id, courses.course_name, students.student_id, students.first_name, students.last_name

    FROM enrollment

    JOIN courses ON enrollment.course_id = courses.course_id

    JOIN students ON enrollment.student_id = students.student_id;

    OUTPUT:

```
MariaDB [revision_worksheet]> SELECT courses.course_id, courses.course_name, students.student_id, students.first_name, students.last_
    -> FROM enrollment
    -> JOIN courses ON enrollment.course_id = courses.course_id
    -> JOIN students ON enrollment.student_id = students.student_id;
+-----------+-----------------------------+------------+-------------+-----------+
| course_id | course_name                 | student_id | first_name  | last_name |
+-----------+-----------------------------+------------+-------------+-----------+
|         1 | Introduction to Programming |          1 | John        | Doe       |
|         3 | Statistics                  |          1 | John        | Doe       |
|         2 | Database Management         |          2 | Jane        | Smith     |
|         4 | English Composition         |          3 | Michael     | Johnson   |
|         5 | History of Art              |          4 | Emily       | Brown     |
|         6 | Chemistry                   |          5 | David       | Martinez  |
|         7 | Economics                   |          6 | Sarah       | Jones     |
|         8 | Algebra                     |          7 | Daniel      | Garcia    |
|         9 | Psychology                  |          8 | Jennifer    | Wilson    |
|        10 | Physical Education          |          9 | Christopher | Anderson  |
+-----------+-----------------------------+------------+-------------+-----------+
10 rows in set (0.001 sec)
```

20. Retrieve students along with the courses they are enrolled in.

    CODING:

    SELECT students.student_id, students.first_name, students.last_name, courses.course_id, courses.course_name

    FROM enrollment

    JOIN students ON enrollment.student_id = students.student_id

    JOIN courses ON enrollment.course_id = courses.course_id;

    OUTPUT:

```
MariaDB [revision_worksheet]> SELECT students.student_id, students.first_name, students.last_name, courses.course_id, courses.course_
    -> FROM enrollment
    -> JOIN students ON enrollment.student_id = students.student_id
    -> JOIN courses ON enrollment.course_id = courses.course_id;
+------------+-------------+-----------+-----------+-----------------------------+
| student_id | first_name  | last_name | course_id | course_name                 |
+------------+-------------+-----------+-----------+-----------------------------+
|          1 | John        | Doe       |         1 | Introduction to Programming |
|          1 | John        | Doe       |         3 | Statistics                  |
|          2 | Jane        | Smith     |         2 | Database Management         |
|          3 | Michael     | Johnson   |         4 | English Composition         |
|          4 | Emily       | Brown     |         5 | History of Art              |
|          5 | David       | Martinez  |         6 | Chemistry                   |
|          6 | Sarah       | Jones     |         7 | Economics                   |
|          7 | Daniel      | Garcia    |         8 | Algebra                     |
|          8 | Jennifer    | Wilson    |         9 | Psychology                  |
|          9 | Christopher | Anderson  |        10 | Physical Education          |
+------------+-------------+-----------+-----------+-----------------------------+
10 rows in set (0.001 sec)
```

21. Retrieve all possible combinations of students and courses.

CODING:

SELECT students.student_id, students.first_name, students.last_name, courses.course_id, courses.course_name

FROM students

CROSS JOIN courses;

OUTPUT:

```
MariaDB [revision_worksheet]> SELECT students.student_id, students.first_name, students.last_name, courses.course_id, courses.course_name
    -> FROM students
    -> CROSS JOIN courses;
+------------+-------------+-----------+-----------+-----------------------------+
| student_id | first_name  | last_name | course_id | course_name                 |
+------------+-------------+-----------+-----------+-----------------------------+
|          1 | John        | Doe       |         1 | Introduction to Programming |
|          2 | Jane        | Smith     |         1 | Introduction to Programming |
|          3 | Michael     | Johnson   |         1 | Introduction to Programming |
|          4 | Emily       | Brown     |         1 | Introduction to Programming |
|          5 | David       | Martinez  |         1 | Introduction to Programming |
|          6 | Sarah       | Jones     |         1 | Introduction to Programming |
|          7 | Daniel      | Garcia    |         1 | Introduction to Programming |
|          8 | Jennifer    | Wilson    |         1 | Introduction to Programming |
|          9 | Christopher | Anderson  |         1 | Introduction to Programming |
|         10 | Amanda      | Thomas    |         1 | Introduction to Programming |
|          1 | John        | Doe       |         2 | Database Management         |
|          2 | Jane        | Smith     |         2 | Database Management         |
|          3 | Michael     | Johnson   |         2 | Database Management         |
|          4 | Emily       | Brown     |         2 | Database Management         |
|          5 | David       | Martinez  |         2 | Database Management         |
|          6 | Sarah       | Jones     |         2 | Database Management         |
|          7 | Daniel      | Garcia    |         2 | Database Management         |
|          8 | Jennifer    | Wilson    |         2 | Database Management         |
|          9 | Christopher | Anderson  |         2 | Database Management         |
|         10 | Amanda      | Thomas    |         2 | Database Management         |
|          1 | John        | Doe       |         3 | Statistics                  |
|          2 | Jane        | Smith     |         3 | Statistics                  |
|          3 | Michael     | Johnson   |         3 | Statistics                  |
|          4 | Emily       | Brown     |         3 | Statistics                  |
|          5 | David       | Martinez  |         3 | Statistics                  |
|          6 | Sarah       | Jones     |         3 | Statistics                  |
|          7 | Daniel      | Garcia    |         3 | Statistics                  |
|          8 | Jennifer    | Wilson    |         3 | Statistics                  |
|          9 | Christopher | Anderson  |         3 | Statistics                  |
|         10 | Amanda      | Thomas    |         3 | Statistics                  |
|          1 | John        | Doe       |         4 | English Composition         |
|          2 | Jane        | Smith     |         4 | English Composition         |
|          3 | Michael     | Johnson   |         4 | English Composition         |
|          4 | Emily       | Brown     |         4 | English Composition         |
|          5 | David       | Martinez  |         4 | English Composition         |
|          6 | Sarah       | Jones     |         4 | English Composition         |
|          7 | Daniel      | Garcia    |         4 | English Composition         |
|          8 | Jennifer    | Wilson    |         4 | English Composition         |
|          9 | Christopher | Anderson  |         4 | English Composition         |
|         10 | Amanda      | Thomas    |         4 | English Composition         |
```

1	John	Doe	4	English Composition
2	Jane	Smith	4	English Composition
3	Michael	Johnson	4	English Composition
4	Emily	Brown	4	English Composition
5	David	Martinez	4	English Composition
6	Sarah	Jones	4	English Composition
7	Daniel	Garcia	4	English Composition
8	Jennifer	Wilson	4	English Composition
9	Christopher	Anderson	4	English Composition
10	Amanda	Thomas	4	English Composition
1	John	Doe	5	History of Art
2	Jane	Smith	5	History of Art
3	Michael	Johnson	5	History of Art
4	Emily	Brown	5	History of Art
5	David	Martinez	5	History of Art
6	Sarah	Jones	5	History of Art
7	Daniel	Garcia	5	History of Art
8	Jennifer	Wilson	5	History of Art
9	Christopher	Anderson	5	History of Art
10	Amanda	Thomas	5	History of Art
1	John	Doe	6	Chemistry
2	Jane	Smith	6	Chemistry
3	Michael	Johnson	6	Chemistry
4	Emily	Brown	6	Chemistry
5	David	Martinez	6	Chemistry
6	Sarah	Jones	6	Chemistry
7	Daniel	Garcia	6	Chemistry
8	Jennifer	Wilson	6	Chemistry
9	Christopher	Anderson	6	Chemistry
10	Amanda	Thomas	6	Chemistry
1	John	Doe	7	Economics
2	Jane	Smith	7	Economics
3	Michael	Johnson	7	Economics
4	Emily	Brown	7	Economics
5	David	Martinez	7	Economics
6	Sarah	Jones	7	Economics
7	Daniel	Garcia	7	Economics
8	Jennifer	Wilson	7	Economics
9	Christopher	Anderson	7	Economics
10	Amanda	Thomas	7	Economics
1	John	Doe	8	Algebra
2	Jane	Smith	8	Algebra
3	Michael	Johnson	8	Algebra
4	Emily	Brown	8	Algebra
5	David	Martinez	8	Algebra
6	Sarah	Jones	8	Algebra
7	Daniel	Garcia	8	Algebra

```
|          7 | Daniel     | Garcia    |          7 | Economics
|          8 | Jennifer   | Wilson    |          7 | Economics
|          9 | Christopher | Anderson  |          7 | Economics
|         10 | Amanda     | Thomas    |          7 | Economics
|          1 | John       | Doe       |          8 | Algebra
|          2 | Jane       | Smith     |          8 | Algebra
|          3 | Michael    | Johnson   |          8 | Algebra
|          4 | Emily      | Brown     |          8 | Algebra
|          5 | David      | Martinez  |          8 | Algebra
|          6 | Sarah      | Jones     |          8 | Algebra
|          7 | Daniel     | Garcia    |          8 | Algebra
|          8 | Jennifer   | Wilson    |          8 | Algebra
|          9 | Christopher | Anderson  |          8 | Algebra
|         10 | Amanda     | Thomas    |          8 | Algebra
|          1 | John       | Doe       |          9 | Psychology
|          2 | Jane       | Smith     |          9 | Psychology
|          3 | Michael    | Johnson   |          9 | Psychology
|          4 | Emily      | Brown     |          9 | Psychology
|          5 | David      | Martinez  |          9 | Psychology
|          6 | Sarah      | Jones     |          9 | Psychology
|          7 | Daniel     | Garcia    |          9 | Psychology
|          8 | Jennifer   | Wilson    |          9 | Psychology
|          9 | Christopher | Anderson  |          9 | Psychology
|         10 | Amanda     | Thomas    |          9 | Psychology
|          1 | John       | Doe       |         10 | Physical Education
|          2 | Jane       | Smith     |         10 | Physical Education
|          3 | Michael    | Johnson   |         10 | Physical Education
|          4 | Emily      | Brown     |         10 | Physical Education
|          5 | David      | Martinez  |         10 | Physical Education
|          6 | Sarah      | Jones     |         10 | Physical Education
|          7 | Daniel     | Garcia    |         10 | Physical Education
|          8 | Jennifer   | Wilson    |         10 | Physical Education
|          9 | Christopher | Anderson  |         10 | Physical Education
|         10 | Amanda     | Thomas    |         10 | Physical Education
+------------+------------+-----------+------------+----------------------------
100 rows in set (0.001 sec)
```

22. Retrieve students who are enrolled in the same course.

CODING:

```
SELECT e1.student_id, e2.student_id AS other_student_id, e1.course_id
FROM enrollment e1
JOIN enrollment e2 ON e1.course_id = e2.course_id AND e1.student_id <> e2.student_id;
```

OUTPUT:

```
MariaDB [revision_worksheet]> SELECT e1.student_id, e2.student_id AS other_student_id, e1.course_
    -> FROM enrollment e1
    -> JOIN enrollment e2 ON e1.course_id = e2.course_id AND e1.student_id <> e2.student_id;
Empty set (0.001 sec)
```

23. Retrieve a combined list of first names from students and courses.

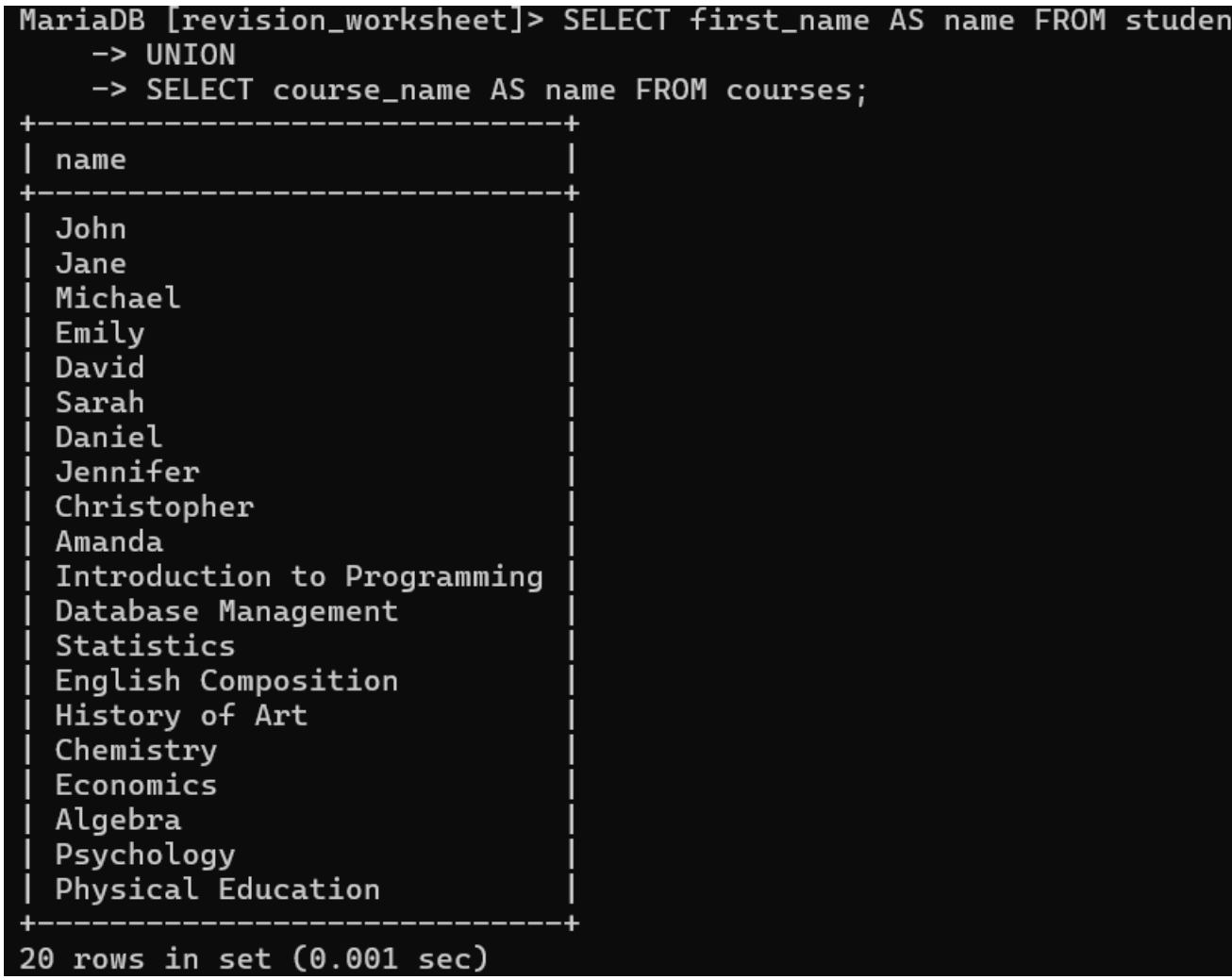
CODING:

```
SELECT first_name AS name FROM students
```

UNION

SELECT course_name AS name FROM courses;

OUTPUT:



```
MariaDB [revision_worksheet]> SELECT first_name AS name FROM studen
    -> UNION
    -> SELECT course_name AS name FROM courses;
+----------------------------+
| name                       |
+----------------------------+
| John                       |
| Jane                       |
| Michael                    |
| Emily                      |
| David                      |
| Sarah                      |
| Daniel                     |
| Jennifer                   |
| Christopher                |
| Amanda                     |
| Introduction to Programming |
| Database Management        |
| Statistics                 |
| English Composition        |
| History of Art             |
| Chemistry                  |
| Economics                  |
| Algebra                    |
| Psychology                 |
| Physical Education         |
+----------------------------+
20 rows in set (0.001 sec)
```